The regular columns of the small-group atlas

Human Compression project

We consider the 79 group models printed in Tables 2–7 of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382. There are 88 table occurrences, since some groups occur more than once. A regular marking of rank r means an ordered tuple of nonidentity involutions $\rho_0, \dots, \rho_{r-1}$ which generates the group, satisfies $\rho_i\rho_j = \rho_j\rho_i$ for $|i-j| > 1$, and has the intersection property

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle \qquad (I, J \subseteq \{0, \dots, r-1\}).$$

Markings are identified by automorphisms carrying each generator to the corresponding generator. Reversal is not additionally identified. The standard correspondence with regular abstract polytopes is retained; we do not assert a classification under a different incidence convention.

Proposition 1. *Relative to the acquired literal models, group-operation implementations, and the reviewed enumeration theorem, the regular columns for all 79 printed groups are determined by the accompanying census files. Of the 88 printed rank-count rows, 87 agree with these results. In Table 4, page 379, the rank-three entry for* $\mathrm{PSL}_2(27)$ *is 26, whereas the source prints 27. All other regular entries agree rank by rank.*

Proof. Fourteen models are supplied by the already reviewed complete regular censuses; their rank lists are acquired without repeating their searches. For the remaining 65 models, the acquired enumeration constructs involutory prefixes up to the complete automorphism action. At an extension, representatives of the relevant double cosets enumerate all inequivalent choices in the required centralizers. The intersection test and the acquired normal-closure rejection remove only impossible prefixes. A string C-group tuple is independent: removing a generator strictly reduces the subgroup it generates. The increasing subgroup chain therefore bounds its rank by $\lfloor \log_2 |G| \rfloor$. These are the hypotheses and conclusion of the previously reviewed enumeration theorem; this application uses that theorem without altering its search or rejection rules.

The action on each literal model is acquired from the reviewed action construction. For a normalizing permutation group A, the equality $|A : C_A(G)| = |\operatorname{Aut} G|$ proves that its conjugation action is complete. Where necessary, the construction instead uses the faithful common action of G and its automorphism group on the involutions. The checked generator map connects that action to the original model. Both constructions and their faithfulness hypotheses are retained inputs.

Every one of the 65 actual executions returned its complete rank list, with a clean terminal result and unchanged input files. The independent receipt verifier reconstructs the complete input, driver and output inventories and reparses the returned census. It neither executes another search nor reads the source counts as search inputs. Joining these 65 lists with the 14 retained lists gives exactly the 79 printed group keys, without overlap. Joining them to all 88 source occurrences gives the stated comparison. The machine-readable result records every rank list, including the source discrepancy and all duplicate joins. □

An independent rank-three check for $\mathrm{PSL}_2(27)$**.** This count also has a short finite verification which does not use the prefix enumeration. Write $G = \mathrm{PSL}_2(27)$ in its natural action on 28 points. Its order is 9828. The accompanying script checks literal equality with the standard constructor, and acquires a normalizing action of order 58968 inducing all automorphisms. There are 351 involutions in G, forming one orbit. Fix an involution x. The script lists all 14 distinct involutions commuting with x and verifies that the stabilizer of x is transitive on them. Thus all possible ordered commuting endpoint pairs (x, z) are equivalent. The stabilizer of one such pair has order 12.

For this fixed pair, the middle generator y must be one of the 351 involutions. Exactly 312 choices generate G together with x, z. For every one of them the explicit subgroup computation gives

$$\langle x, y \rangle \cap \langle y, z \rangle = \langle y \rangle.$$

This is the rank-three intersection criterion: the rank-two subgroups are dihedral string C-groups, and their intersection is the common rank-one subgroup. Equivalently, the other rank-two intersections follow from $\langle x, z\rangle \cong C_2^2$, the displayed equality, and generation. The endpoint stabilizer acts freely on the 312 possibilities, because an automorphism fixing the three generating elements is the identity. Consequently the number of rank-three markings is $312/12 = 26$. The script separately checks its orbits and their sizes. Enumeration of all elements, subgroup operations and the automorphism-group order are explicit GAP premises of this independent finite check. This paragraph makes no assertion about higher ranks on its own.

Scope and attribution. The source determines the group roster and the marking convention. The previous complete regular censuses, the enumeration theorem and the common-action construction supply the reused mathematics. The new work here is their complete application to the 65 remaining models, the join to all printed regular entries, and the separate fixed-endpoint check of the exceptional row. This is a correction of a finite published count, not a claimed solution of an open conjecture or a claim of priority. The chiral columns, first atlas, and the scope of the historical ATLAS roster require their separate arguments. No whole-manuscript completion, Lean verification or compression measurement follows from this component.